SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Operating Systems **COURSE CODE: CSA04**

COURSE OUTCOME COVERED

CO5: *Analyze the components and functionalities of Linux operating systems and virtualization environments to enhance their operational efficiency*

Assessment Tool Weightage: 30%

QUESTIONS: 5 **Total Marks:25**

Annexure A

MOCK INTERVIEW

Code of Conduct:

I swetha.v (192524180) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

swetha.v

Signature of the student:

Annexure A

MOCK INTERVIEW

S.No.	Questions	Mark
1	How would you analyze the Linux file system structure and implement a long-term solution to prevent this issue?	3
2	You are given a 10 GB log file and need to extract all failed login attempts within 1 minute. Analyze the scenario and propose suitable Linux commands or shell techniques, along with performance optimization strategies, to accomplish the task efficiently.	5
3	A system is experiencing performance degradation due to a process consuming 95% of CPU resources. Analyze the situation and explain how you would identify, evaluate, and manage the process using appropriate Linux tools, without disrupting critical services.	5
4	Would you recommend a Type 1 or Type 2 hypervisor? Justify your answer based on performance, cost, and scalability.	3
5	A server with 16 GB of RAM must support multiple virtual machines. Analyze the scenario and propose an efficient resource allocation strategy to ensure optimal performance and balanced utilization across all VMs.	4
6	A Linux server runs multiple applications, where one application is negatively impacting the performance of others. Analyze the situation and propose a resource isolation strategy to ensure fair and controlled usage of CPU and memory among all applications.	5

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically	Provides reasonable	Superficial or partially structured	No analytical thinking

		and provides well-structured, logical answers	analysis with some structure	responses	demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Respondsaccurately,promptly,andelaborateswith Respondscorrectlybutwithlimitedd Partialresponses;needspromptin Unabletorespondappropriately	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately

relevant examples pth g					

1. How would you analyze the Linux file system structure and implement a long-term solution to prevent this issue?

Answer:

I would first analyze the Linux file system using commands such as:

df -h

du -sh /*

lsblk

df -h shows the disk space used by each file system, while du -sh helps identify which directories are consuming the most space. lsblk displays the available disks and partitions.

I would then identify large files, especially log files, temporary files, and unnecessary packages. For a long-term solution, I would configure **log rotation using logrotate**, regularly remove unnecessary files, monitor disk usage, and set alerts when disk utilization becomes high. This prevents the file system from becoming full and affecting system performance.

2. You are given a 10 GB log file and need to extract all failed login attempts within 1 minute. Analyze the scenario and propose suitable Linux commands or shell techniques, along with performance optimization strategies, to accomplish the task efficiently.

Answer:

Since the log file is very large, I would avoid opening the entire file in an editor. I would use command-line tools such as grep, awk, and sed because they can process the file efficiently.

For example, if the log contains entries such as Failed password, I can search for failed login attempts using:

grep "Failed password" /var/log/auth.log

To filter a particular one-minute time interval, I can use awk based on the timestamp.

For example:

awk '$0 >= "10:15:00" && $0 <= "10:16:00" && /Failed password/' /var/log/auth.log

For a 10 GB file, performance can be improved by:

1. Using grep instead of opening the file in a text editor.

2. Searching only the required time range.
3. Using awk to filter records in a single pass.
4. Using grep -F when searching for a fixed string.
5. Processing the file sequentially instead of loading the complete file into memory.
6. Using compressed logs with zgrep if the log has already been compressed.

Therefore, command-line filtering provides an efficient way to process large log files with minimal memory usage.

3. A system is experiencing performance degradation due to a process consuming 95% of CPU resources. Analyze the situation and explain how you would identify, evaluate, and manage the process using appropriate Linux tools, without disrupting critical services.

Answer:

First, I would identify the CPU-consuming process using:

top

or:

ps aux --sort=-%cpu | head

These commands show which process is consuming the highest CPU resources. I would note its **PID, CPU usage, memory usage, and process name**.

Next, I would evaluate whether the process is a critical system service or an unnecessary process. I would also check its behavior using:

top -p PID

If the process is non-critical, I can reduce its CPU priority using:

renice 10 -p PID

If it is a critical process, I would avoid immediately terminating it. Instead, I would investigate the application logs and identify the reason for excessive CPU consumption.

If the process becomes completely unresponsive and is confirmed to be safe to terminate, I can use:

kill PID

If necessary, a stronger signal can be used carefully:

kill -9 PID

However, kill -9 should be the last option because it does not allow the application to perform normal cleanup.

Therefore, I would **identify → evaluate → reduce priority or control resources → monitor**, while avoiding disruption to critical services.

4. Would you recommend a Type 1 or Type 2 hypervisor? Justify your answer based on performance, cost, and scalability.

Answer:

I would recommend a **Type 1 hypervisor** for production and enterprise environments. A Type 1 hypervisor runs directly on the physical hardware without requiring a traditional host operating system. Therefore, it generally provides better performance and efficient resource utilization.

Factor	**Type 1**	**Type 2**
Performance	High	Moderate
Cost	Usually higher for enterprise deployment	Generally lower/easier for desktop use
Scalability	Excellent	Limited compared with Type 1
Usage	Servers and data centers	Development and testing

Examples of Type 1 hypervisors include **VMware ESXi, Microsoft Hyper-V, and Xen**.

Therefore, for a system requiring **high performance, scalability, reliability, and efficient resource utilization**, I would choose Type 1.

5. A server with 16 GB of RAM must support multiple virtual machines. Analyze the scenario and propose an efficient resource allocation strategy to ensure optimal performance and balanced utilization across all VMs.

Answer:

I would first reserve sufficient memory for the host operating system because the host also needs RAM to perform its own operations.

For example:

Total RAM = 16 GB

If I reserve:

Host OS = 4 GB

Then:

RAM available for VMs = 16 − 4 = 12 GB

If there are **4 VMs**, I could initially allocate:

12 ÷ 4 = 3 GB per VM

So the allocation would be:

Resource	Allocation
Total RAM	16 GB
Host OS	4 GB
VM 1	3 GB
VM 2	3 GB
VM 3	3 GB
VM 4	3 GB

I would monitor RAM usage using tools such as free, top, or virtualization management tools. If one VM requires more memory while another is underutilized, resources can be adjusted according to workload.

I would also avoid allocating all physical RAM to VMs because the host requires memory for management and other services.

Therefore, **reserving host memory, allocating minimum required RAM, monitoring utilization, and dynamically adjusting resources** provides balanced performance.

6. A Linux server runs multiple applications, where one application is negatively impacting the performance of others. Analyze the situation and propose a resource isolation strategy to ensure fair and controlled usage of CPU and memory among all applications.

Answer:

I would use **Linux control groups (cgroups)** to isolate and control the resources used by each application.

First, I would identify the application consuming excessive resources using:

top

or:

ps aux --sort=-%cpu

Then I would create a separate cgroup for the application and configure CPU and memory limits.

For example, conceptually:

```
Application A
   ↓
Cgroup A
   ├── CPU limit
   └── Memory limit

Application B
   ↓
Cgroup B
   ├── CPU limit
   └── Memory limit
```

The CPU limit prevents one application from continuously consuming most of the CPU. The memory limit prevents it from consuming excessive RAM and affecting other applications.

I would then monitor resource usage and adjust the limits according to the workload. For example, a critical application can be given higher CPU and memory limits, while a background application can receive lower limits.

Resource Isolation Strategy

1. Identify the resource-consuming application.
2. Create a separate cgroup for the application.
3. Set CPU limits.
4. Set memory limits.
5. Assign the application's processes to the cgroup.
6. Monitor CPU and memory usage.
7. Adjust limits when necessary.

This ensures **fair resource allocation, process isolation, better stability, and improved overall server performance**.